

# Stochastic Contextual Bandits / Part 1

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# $K$ -armed stochastic bandits

Simplest possible framework

$K$  probability distributions  $\nu_1, \dots, \nu_K$  in a model  $\mathcal{D}$   
with expectations  $\mu_1, \dots, \mu_K \longrightarrow \mu^\star = \max_{a \in [K]} \mu_a$

At each round  $t = 1, 2, \dots$ ,

1. Statistician picks **arm**  $A_t \in [K]$  (possibly at random)
2. She gets a reward  $Y_t$  drawn according to  $\nu_{A_t}$
3. This is the **only feedback** she receives

$\longrightarrow$  **Exploration–exploitation dilemma**  
estimate the  $\nu_a$  **vs.** get high rewards  $Y_t$

Goal:

Maximize expected cumulative rewards  $\longleftrightarrow$  Minimize **regret**

$$R_T = T\mu^\star - \mathbb{E}\left[\sum_{t=1}^T Y_t\right] = \sum_{a \in [K]} (\mu^\star - \mu_a) \mathbb{E}[N_a(T)]$$

$\longleftrightarrow$  Control the  $\mathbb{E}[N_a(T)]$  for suboptimal arms  $a$

Setting:

Distributions  $\nu_1, \dots, \nu_K$  with expectations  $\mu_1, \dots, \mu_K$

At each round  $t \geq 1$ , pick arm  $A_t \in [K]$ , get and observe  $Y_t \sim \nu_{A_t}$

## Proof of the rewriting of regret

Tower rule:  $\mathbb{E}[Y_t | A_t] = \mu_{A_t}$       thus       $\mathbb{E}[Y_t] = \mathbb{E}[\mu_{A_t}]$

$$\begin{aligned} R_T &= \sum_{t=1}^T (\mu^* - \mathbb{E}[Y_t]) = \sum_{t=1}^T (\mu^* - \mathbb{E}[\mu_{A_t}]) \\ &= \sum_{t=1}^T \sum_{a \in [K]} (\mu^* - \mu_a) \mathbb{E}[\mathbb{I}_{\{A_t=a\}}] = \sum_{a \in [K]} (\mu^* - \mu_a) \mathbb{E}[N_a(T)] \end{aligned}$$

where 
$$N_a(T) = \sum_{t=1}^T \mathbb{I}_{\{A_t=a\}}$$

## Noise assumption: $\sigma^2$ -sub-Gaussian

$$Y_t \sim \nu_{A_t} \quad \text{made more precise:} \quad Y_t = \mu_{A_t} + \eta_t$$

Noise  $\eta_t$  independent from “everything else” and  $\sigma^2$ -sub-Gaussian:

$$\forall \lambda \in \mathbb{R}, \quad \mathbb{E}[e^{\lambda \eta_t} \mid \mathcal{F}_{t-1}] = \mathbb{E}[e^{\lambda \eta_t}] \leq e^{\lambda^2 \sigma^2 / 2}$$

$\mathcal{F}_{t-1}$  represents the information:  $Y_1, \dots, Y_{t-1}$  and  $A_1, \dots, A_t$

In particular,  $\mathbb{E}[e^{\lambda \eta_t} \mid \mathcal{F}_{t-1}] = \mathbb{E}[e^{\lambda \eta_t}] = 0$  [proof by Taylor expansions]

Examples:

- $\mathcal{N}(0, \sigma^2)$  noise
- noise taking values in  $[a, b]$  is  $(b - a)^2/4$ -sub-Gaussian (cf. Hoeffding's lemma)

# Explore-then-commit strategy

As a warm-up

## Explore-then-commit strategy

Reference: e.g., Lattimore and Szepesvári [2020, Chapter 6]

### 1. Explore for $mK$ rounds

Pull each arm  $m$  times, compute empirical means  $\hat{\mu}_a$  for  $a \in [K]$

### 2. Exploit for $T - mK$ rounds

For  $t \geq mK + 1$ , pull an arm  $A_t \in \arg \max_{a \in [K]} \hat{\mu}_a$

Regret bound (under the  $\sigma^2$ -sub-Gaussian noise assumption):

$$R_T \leq \underbrace{m \sum_{a \in [K]} (\mu^* - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^* - \mu_a) \exp\left(-\frac{m(\mu^* - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

## Hoeffding's inequality

Reminder:  $\eta$  is  $\sigma^2$ -sG if  $\mathbb{E}[e^{\lambda\eta}] \leq e^{\lambda^2\sigma^2/2}$

If  $\eta, \eta'$  are  $\sigma^2$ -sG and independent, then  $(\eta + \eta')/2$  is  $\sigma^2/2$ -sG

Cramér-Chernoff: for  $\lambda = \varepsilon/\sigma^2 > 0$ ,

$$\mathbb{P}(\eta \geq \varepsilon) \leq e^{-\lambda\varepsilon} \mathbb{E}[e^{\lambda\eta}] \leq \exp(-\lambda\varepsilon + \lambda^2\sigma^2/2) = \exp(-\varepsilon^2/(2\sigma^2))$$

**Hoeffding's inequality:**  $\eta_1, \dots, \eta_m$  independent and  $\sigma^2$ -sG

$$\mathbb{P}\left(\frac{1}{m} \sum_{t=1}^m \eta_t \geq \varepsilon\right) \leq \exp(-m\varepsilon^2/(2\sigma^2))$$

$$\text{or} \quad \text{w.p. } 1 - \delta, \quad \frac{1}{m} \sum_{t=1}^m \eta_t \leq \sigma \sqrt{\frac{2}{m} \ln(1/\delta)}$$

## Back to explore-then-commit

Regret bound (with  $\sigma^2$ -sub-Gaussian noises):

$$R_T = \sum_{t=1}^T (\mu^* - \mathbb{E}[\mu_{A_t}]) \leq \underbrace{m \sum_{a \in [K]} (\mu^* - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^* - \mu_a) \exp\left(-\frac{m(\mu^* - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Exploitation costs  $\longleftrightarrow a^* \in \arg \max_{a \in [K]} \mu_a \neq \arg \max_{a \in [K]} \hat{\mu}_a$

$$\mathbb{P}(\hat{\mu}_a \geq \hat{\mu}_{a^*}) = \mathbb{P}\left(\frac{1}{m} \sum_{t=1}^m \underbrace{\eta_t - \eta'_t}_{2\sigma^2\text{-sG}} \geq \mu^* - \mu_a\right) \leq (-m\varepsilon^2/(4\sigma^2))$$

where  $\varepsilon = \mu^* - \mu_a$

## Two types of regret bounds

Gap-dependent regret bound:

$$R_T = \sum_{t=1}^T (\mu^* - \mathbb{E}[\mu_{A_t}]) \leq \underbrace{m \sum_{a \in [K]} (\mu^* - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^* - \mu_a) \exp\left(-\frac{m(\mu^* - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Involves the gaps  $\Delta_a = \mu^* - \mu_a$

More generally called: problem-dependent bound

Now, uniform bound (**gap-free bound** or worst-case bound):

in a model  $\mathcal{D}_{\sigma^2, B}$  with  $\Delta_a \leq B$

Using  $xe^{-x^2} \leq 1/2$  for  $x \geq 0$

$$\sup_{\mathcal{D}_{\sigma^2, B}} R_T \leq mKB + (T - mK) \frac{2\sigma}{\sqrt{m}} \frac{1}{2} \leq mKB + \sigma \frac{T}{\sqrt{m}} = \mathcal{O}(T^{2/3})$$

for  $m \sim T^{2/3}$

# Upper-confidence-bound [UCB] strategy

“Optimism in the face of uncertainty”

Model:  $\nu_1, \dots, \nu_K$  are  $\sigma^2$ -sG

A popular strategy: **UCB** [upper confidence bound]

Auer, Cesa-Bianchi and Fisher [2002]

Pull each arm once

For  $t \geq K$ , pick  $A_{t+1} \in \arg \max_{a \in [K]} \left\{ \hat{\mu}_a(t) + \sigma \sqrt{\frac{8 \ln t}{N_a(t)}} \right\}$

**Exploitation**: cf. empirical mean  $\hat{\mu}_a(t) = \frac{1}{N_a(t)} \sum_{s=1}^t Y_s \mathbb{I}_{\{A_s=a\}}$

**Exploration**: cf.  $\sigma \sqrt{8 \ln t / N_a(t)}$  favors arms  $a$  not pulled often

Regret bounds (slightly suboptimal) of two types

– **Gap-dependent** bound:  $R_T \lesssim \square \sum_{a: \mu_a < \mu^*} \frac{\sigma \ln T}{\mu^* - \mu_a}$

– **Gap-free** bound:  $\sup \sigma^2 R_T \lesssim \square \sigma \sqrt{KT \ln T}$   
 $\sigma^2$ -sG

Proof of  $R_T \lesssim \square \sum_{a: \mu_a < \mu^*} \frac{\sigma \ln T}{\mu^* - \mu_a}$

Hoeffding–Azuma ext.:  $\mathbb{P}\left(|\mu_a - \hat{\mu}_a(t)| \leq \sigma \sqrt{\frac{8 \ln t}{N_a(t)}}\right) \geq 1 - 2t^{-3}$

Recall  $\hat{\mu}_a(t) = \frac{1}{N_a(t)} \sum_{s=1}^t Y_s \mathbb{I}_{\{A_s=a\}}$

Indeed, by **Doob's optional skipping**:

$$\begin{aligned} & \mathbb{P}\left(|\mu_a - \hat{\mu}_a(t)| > \sigma \sqrt{\frac{8 \ln t}{N_a(t)}}\right) \\ &= \sum_{n=1}^t \mathbb{P}\left(|\mu_a - \hat{\mu}_{a,n}| > \sigma \sqrt{\frac{8 \ln t}{n}} \quad \text{and} \quad N_a(t) = n\right) \\ &\leq \underbrace{\sum_{n=1}^t \mathbb{P}\left(|\mu_a - \hat{\mu}_{a,n}| > \sigma \sqrt{\frac{2 \ln(1/t^{-4})}{n}}\right)}_{\leq 2t^{-4}, \quad \text{cf. classic Hoeffding-Azuma applied twice}} \end{aligned}$$

where  $\hat{\mu}_{a,n}$  denotes a  $n$ -sample average with distribution  $\nu_a$

Proof of  $R_T \lesssim \square \sum_{a: \mu_a < \mu^*} \frac{\sigma \ln T}{\mu^* - \mu_a}$

Hoeffding–Azuma ext.:  $\mathbb{P}\left(|\mu_a - \hat{\mu}_a(t)| \leq \sigma \sqrt{\frac{8 \ln t}{N_a(t)}}\right) \geq 1 - 2t^{-3}$

If  $A_t = b$  is not an optimal arm  $a^*$ , then by design

$$\hat{\mu}_{a^*}(t) + \sigma \sqrt{\frac{8 \ln t}{N_{a^*}(t)}} \leq \hat{\mu}_b(t) + \sigma \sqrt{\frac{8 \ln t}{N_b(t)}}$$

thus w.p.  $\geq 1 - 4t^{-3}$ ,  $\mu^* \leq \mu_b + 2\sigma \sqrt{\frac{8 \ln t}{N_b(t)}}$

which imposes  $N_b(t) \leq \frac{32\sigma \ln T}{(\mu^* - \mu_b)^2}$

$\mathbb{E}[N_b(T)]$  controlled by this  $+ \square t^{-2}$

Conclude with  $R_T = \sum_{a \in [K]} (\mu^* - \mu_a) \mathbb{E}[N_a(T)]$

## Proof of $\sup_{\sigma^2\text{-sG}} R_T \lesssim \sigma \sqrt{KT \ln T}$

We proved  $\mathbb{E}[N_b(t)] \lesssim \square \sigma \frac{\ln T}{(\mu^* - \mu_b)^2}$

Thus 
$$\begin{aligned} R_T &= \sum_{a \in [K]} (\mu^* - \mu_a) \sqrt{\mathbb{E}[N_a(T)]} \sqrt{\mathbb{E}[N_a(T)]} \\ &\leq \square \sigma \sqrt{\ln T} \sum_{a \in [K]} \sqrt{\mathbb{E}[N_a(T)]} \\ &\leq \square \sigma \sqrt{KT \ln T} \end{aligned}$$

cf. Jensen for  $\sqrt{\cdot}$

# Thompson sampling

A Bayesian view

## Thompson sampling: a Bayesian view

Studied mostly for exponential families  $\mathcal{D} \leftrightarrow \Theta$

E.g., Bernoulli  $\Theta = (0, 1)$ ; Gaussian  $\Theta = \mathbb{R}$  with known variance  $\sigma^2$

Ref.: Thompson [1933]; Agrawal and Goyal [2012, 2013, 2017];

Kaufmann, Korda and Munos [2012, 2013]

Well-chosen prior distribution  $\pi$  over  $\Theta$

$\mathcal{D}$  Bernoulli  $\rightarrow \pi$  Beta;  $\mathcal{D}$  Gaussian  $\rightarrow \pi$  Gaussian

Posterior distributions:  $\hat{\pi}_{t,a}$  for arm  $a \in [K]$

Based on  $\pi$  and rewards  $Y_s$  such that  $1 \leq s \leq t-1$  and  $A_s = a$

**Strategy:** draw  $\theta_{t,a} \sim \hat{\pi}_{t,a}$  <sup>exploration</sup> and pick  $A_t \in \arg \max_{a \in [K]} \mu(\theta_{t,a})$  <sup>exploitation</sup>

## Thompson sampling: Bernoulli case $\Theta = (0, 1)$

Distributions indexed by  $p_1, \dots, p_K$  with best parameter  $p^\star$

Prior  $\pi = \text{Beta}(1, 1) = \text{uniform distribution over } (0, 1)$

Posteriors:  $\hat{\pi}_{t,a} = \text{Beta}(1 + S_a(t), 1 + N_a(t) - S_a(t))$

$$\text{where} \quad N_a(t) = \sum_{s \leq t} \mathbb{I}_{\{A_s = a\}} \quad \text{and} \quad S_a(t) = \sum_{s \leq t} Y_s \mathbb{I}_{\{A_s = a\}}$$

Regret bounds (complex proofs)

$$R_T \lesssim \sum_{a: p_a < p^\star} \frac{\ln T}{\text{kl}(p_a, p^\star)} \leq \sum_{a: p_a < p^\star} \frac{\ln T}{2(p^\star - p_a)^2}$$

thus also  $R_T \lesssim \square \sqrt{KT \ln T}$